

Food based dietary guidelines and diet planning



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Fundamentals of Human Nutrition

Learning objectives

- Identify the Sri Lankan dietary guidelines
- Plan a healthy diet using dietary guidelines for different stages of life

Food based dietary guidelines (FBDG) definition

- Science-based recommendations in the form of guidelines for healthy eating.
- For consumer information
- Appropriate for the region or country
- Each country should have FBDGs
- Culturally acceptable and practical to implement

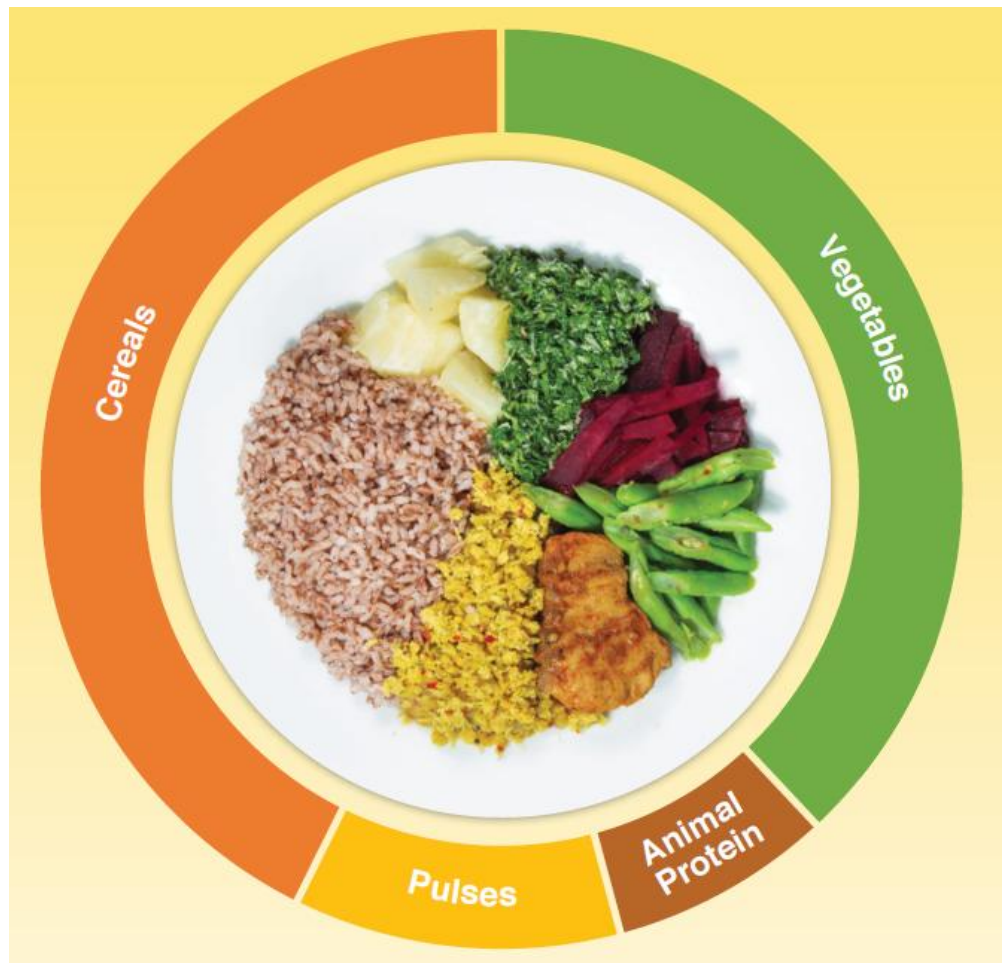
FBDG for Sri Lankans

1. Add colour to your daily meals, balancing the correct amounts

- Incorporate following food groups

- a. Cereals and starchy foods
- b. Fruits
- c. Vegetables and green leaves
- d. Fish/meat/eggs/pulses
- e. Milk and milk products
- f. Nuts, oil and oily seeds





1. Serve half of the plate with cereals and starchy food (parboiled or less polished rice and boiled or curried yams/jack fruit/breadfruit as per preference). Some amount of carbohydrate will be provided from pulses too.
2. Fill approximately $\frac{2}{3}$ of other half of the plate with at least 2 vegetables and one green leafy vegetable.
3. Fill the rest of the plate ($\frac{1}{3}$ of the other half) with protein sources food. Out of which $\frac{2}{3}$ should be from plant sources of protein and $\frac{1}{3}$ from animal sources of protein.

2. Eat whole grains instead of refined

- Less wheat flour preparations
- Identify the serving sizes

Cereals & starchy food - mainly provide carbohydrate
8-13 servings a day



Cooked rice
½ cup (65 g)



String hoppers - made
with rice flour
2 medium size (10 g each)



Bread
1 slice (30 g)



1/4 of a large pittu (3 cm height & 5 cm diameter)



1 hopper (25 g)



1/2 of a roti (about 9 cm diameter & 0.5 cm thick)



1 medium size potato (100g)



1 dosai (20-25g)



Boiled sweet potato/ manioc/
other yams/ jackfruit/ breadfruit
1/2 cup

3. Eat two vegetables one leafy green vegetable and two fruits daily

- Eat underutilized fruits and vegetables too

Vegetables, green leaves and fruits - mainly provide vitamins and minerals

Vegetables and green leaves - 3-5 servings per day



Vitamin A rich vegetables - e.g. pumpkin, carrot, butternut squash

1 serving size = $\frac{1}{2}$ cup or 3 tbsp



Other vegetables including root vegetables - e.g. beans, long beans, snake gourd, ridge gourd, drumstick, bitter gourd, brinjal, banana flower, capsicum, wing beans, tomato, cauliflower, *Amberella*, beet, radish, knol-khol, lasia, lotus roots



Green leaves - e.g. *Kathurumurunga*, drumstick leaves, *Thampala*, *Saarana*, *Mukunuwenna*, *Gotukola*, *Kankun*, *Anguna*, *Thebu*, beet leaves, radish leaves, pumpkin leaves, spinach, carrot leaves, knol-khol leaves, cabbage leaves, passion leaves, manioc leaves, tender *Kohila* leaves, onion leaves



1 serving size = $\frac{1}{2}$ cup or 3 tbsp

Fruits - 2-3 servings a day



1 medium banana



1/2 small bael fruit



4 tsp raisins

high
carb fruits



1/2 cup Anoda



2 dates



2 cups cubed papaya



1-2 small guava



2 slices pine apple



1 medium mango



1 small orange



1 small wood apple

medium
carb fruits



1 small apple



1 medium
pomegranate



5-10 grapes



10-15 jambu



1 small pear



Veralu, Lovi,
Uguressa,
Dan, lemon,
Nelli,
water melon

low carb fruits
no need to
restrict

1/3 medium avocado (50 g) - rich in healthy fat



4. Add fish, eggs or lean meat together with pulses in each meal

- Reduce the consumption of red meat (ex – beef, pork)
- Remove skin part in chicken
- Vegetarians : consume more pulses



Pulses, fish, egg, lean meat, milk - mainly provide proteins

Pulses - 3-5 serving per day



Cooked dhal	3 tbsp
Cooked <i>kadala parippu</i>	3 tbsp
Cooked <i>mung parippu</i>	3 tbsp

Boiled chickpea	½ cup (75 g)
Boiled cowpea	½ cup 75 g)
Boiled green gram	½ cup (75 g)
Boiled soyabean	½ cup 75 g)

1 serving = ½ cup or 3 tbsp

Fish/lean meat - 2-4 servings per day



Fish	30 g
Chicken	30 g
Beef	30 g
Pork	30 g
Mutton	30 g
Dried sprats	15 g (9 – 10 sprats)
Dried fish	15 g (one match box size piece)

1 serving = 30 g (two match box size piece / 2cm x 3 cm x 4 cm)

Egg - 1 per day



1 serving = 1 egg

5. Consume fresh milk or fermented milk products

Fresh milk and its fermented products - 1/2-1 servings per day



Fresh milk
1 cup (200 ml)



Curd
 $\frac{1}{2}$ cup (100 g)



Yoghurt
1 cup (100g)



Cheese
15 g

6. Eat handful of nuts, oily seeds and oil daily (ex – avocado, peanut, sesame)

Additional guidelines

- Limit : deep fried foods, red meat, butter
- Limit trans fats : deep fried foods, margarine, processed foods
- Don't use oil repeatedly : will form trans fat



Nuts, Oily Seeds and Oils - mainly provide fat



Nuts and oily seeds

2 servings per day

1 tbsp full of peanuts

7 cashews

1 tbsp gingelly seeds (15g)



Oils - 1-3 servings per day

*1 serving = 1 tsp (about 5g)



Coconut (scrapped/kernel/milk)

3-6 servings per day

*1 serving = 1 tbsp (15g)



Tea cup = 200 ml



1 table spoon
(tbsp) = 15 ml/g



1 tea spoon
(tsp) = 5ml/g



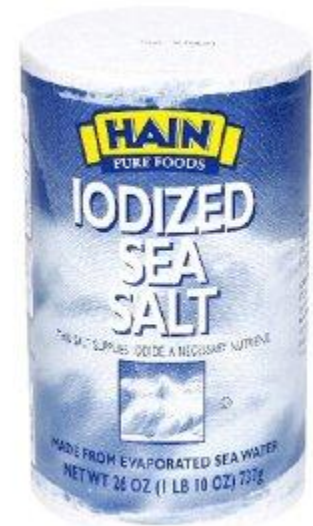
1 match box size =
4 cm x 3 cm x 1 cm

7. Limit salt intake & use iodized salt

- 5 g per day (2300 mg of Na)

Eg:

pickles, ready to eat meals, processed meats (eg: sausages, bacon, ham and salami etc.), cheese, stock cubes, sauces (eg: soy sauce, ketchup etc.), salty snacks, instant noodles, chips, salted dry roasted nuts, some bread and processed cereal products.



8. Eat less sugar and sweet foods

- Those are empty calorie foods
- Risk for diabetes



9. Drink 8 – 10 glasses of water

$$\text{Requirement of fluid (L)} = \frac{\text{Body Weight (kg)}}{30}$$



1

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2



3



4

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10.Be physically activitive

Be active to sweat at least for half an hour a day for 5 days a week.

Physical Activity Recommendations



Moderate Intensity Exercises

5 - 17 years - At least an average of 60 minutes daily across the week

18 - 64 years - Five days a week at least 150-300 minutes per week

65 years and above - Five days a week, at least 150 - 300 minutes per week

Pregnancy and post partum period - At least 150 minutes through out the week

Serving sizes and types of foods for each stage of life

Pregnancy

Guidelines on weight gain in pregnancy

BMI (kg/m ²)		Expected weight gain in kg
Underweight	<18.5	12.5-18
Normal	18.5-24.9	11.5-16
Overweight	25-29.9	7.0-11.5
Obese	>= 30	< 6.8

Source - Maternal care package - A Guide to Field Healthcare Workers, Sri Lanka, 2011 (10)

Recommended number of servings from each food group during pregnancy - 1900kcal for a sedentary and 2500kcal for very active women.



Cereals and starchy food
(8-10 servings)



Pulses (3 servings);
Fish/egg/lean meat (3-5 servings)
An egg (1 serving)



Vegetables and
green leaves (4-5 servings)



Fruits
(3-4 servings)



Fresh milk and its fermented
products ($\frac{1}{2}$ - 1 servings)



Nuts, oil seeds and
oils (2 servings)

Serving sizes and types of foods for each stage of life

Child less than 6 months : Breast milk

Working mother

- Mothers are recommended to express breastmilk which can be given to the baby during her absence by the caregiver.
- Expressed breastmilk can be kept in a clean covered container for 4 - 6 hours at room temperature and for 24 hours in the refrigerator (non-freeze compartment).
- Do not boil the expressed breastmilk before feeding.
- Expressed breastmilk should be fed with a cup or a cup and spoon.
- **Training the mother and the caregiver should start well ahead, at least 4 weeks before going back to work. The third postpartum home visit by the public health midwife (PHM) is a good opportunity to train/support the mother to express breast milk. completion of 4 months after seeking medical advice.**
- **If the mother has to be away at work for long periods and is faced with difficulty in expressing and storing breast milk, still may not need infant formula milk. Complementary feeding can be started at the completion of 4 months after seeking medical advice.**

- 6 – 12 months of age

Guide to correct complementary feeding.



1. Rice is the best food to begin with.

It should be mashed and of semisolid consistency and gradually the consistency should be increased. Pulses, food of animal origin (fish/meat etc), vegetables, green leaves and fruits should be added



2. Introduce foods of animal origin & pulses (preferably germinated) early.

Fish, sprats, poultry, meat, liver– which are rich sources of haem iron. Include green gram, cowpea, *kadala*, soya, *thampala*, *sarana*, *gotukola* alternatively in the daily diet (provide non-haem iron which is not well absorbed as haem iron).



3. Add one or more out of liver, egg yolk, yellow pumpkin, yellow sweet potato, carrot, papaw, mango & dark green leaves - which are rich sources of Vitamin A.



4. Include locally available fruits daily

Fruits should be given in the mashed or pulped form initially and not in the form of juices.



5. Give food with adequate amount of fats.

Add coconut milk or coconut oil when cooking food or fat spreads or butter to the cooked food - this makes food softer, palatable, tastier and energy dense as well as facilitate absorption of fat soluble vitamins.



6. Eggs are a cheap source of good quality protein and can be given daily.

Eggs, a very nutritious food can be added to the baby's diet around 2 weeks after introducing complementary food, starting with egg yolk. Egg should be given well cooked (either boiled or fried until firm). Egg white can be introduced a few days after introducing egg yolk.



7. For a breastfed child other milk/milk based products are not necessary. For a non breastfed child milk/milk based products such as yoghurt and curd can be given as snacks from seventh month.
Curd should be given without adding sugar or honey.



8. Give locally available, low cost nutritious food (rich in energy, proteins, vitamins and minerals) than processed, packeted food.



9. **Do not add sugar or salt in the preparation of food during the first year.** Most natural foods contain adequate amounts of salt & sugar. Taste for added salt & sugar is acquired and should not be encouraged even after the first year.



10. **Avoid foods that can choke child.** Eg. small, rounded, hard pieces of cooked vegetables, Whole nuts/pulses which are hard in consistency.



11. **Ensure variety in every meal.**



12. **Avoid blending or straining of food after 8 months.** Consistency of complementary food should be gradually increased from well mashed at the very beginning to normal adult food by one year of age.



13. **Start feeding with cup and spoon and later switch to fingers according to the consistency of food.**

Number of servings per day from different types of food for children aged 6 months to 5 years

Food group	6+ – 8 months	9-11 months	1+2 years	2+5 years
 Cereals and starchy food	2-3	3-4	3-4	4 or more
 Vegetables	1	1-2	2	2
 Green leaves	1	1	1	1
 Lentils, pulses, nuts	1	1	1-2	1-2
 Fish, meat and egg	1	1-2	1-2	2
 Fruits	1	1-2	2	2
 Breast milk	After meals as desired	After meals as desired	2-3 times per day after meals	2-3 times per day after meals
 Milk products (e.g. curd, yoghurt, cheese) not essential for a breastfed baby	½-1	1	1	1-2
 In case of formula fed children	500-600ml	500-600ml	2 cups (400ml)	2 cups (400ml/d)
 Fat based food	1 tsp/meal	1-2 tsp/meal	Small amount	Small amount
 Confectionary (Not to be encouraged even after the first year)	Should not be given	Should not be given	A small amount after mealtime	A small amount after mealtime
 Salt	Should not be given	Should not be given	Use sparingly	Use sparingly

Give infants water after each meal and whenever they are thirsty using a cup.

Children for 6 – 10 years

Number of Servings for food groups recommended for children 6 – 10 years old (1400 kcal – 1900 kcal).

Food	Servings per day
 Cereals and starchy food	6 - 8
 Vegetables and green leaves	3
 Pulses	2 - 3
 Fish, lean meat	2 - 3
 Egg	1
 Fruits	2 - 3
 Fresh milk and its fermented products	1 - 1 ½
 Nuts and oily seeds	1
 Oils	1 - 2
 Coconut, scraped/milk	3 - 6

Children and adolescents

Number of servings of each food group recommended for adolescents (11 – 18 year old). Based on 2000 kcal - 3400 kcal energy requirement for 11-18 year old adolescents.

Food	Servings per day
 Cereals and starchy food	11 - 18
 Vegetables and green leaves	4-5
 Pulses	3 - 4
 Fish, lean meat	3 - 5
 Egg	1
 Fruits	2 - 4
 Fresh milk and its fermented products	1 - 2*
 Nuts and oily seeds	1 - 3
 Oils	2 - 4
 Coconut, scraped/milk	3 - 6